

Feasible regions



1 The set of a ordered pairs that satisfy each inequality in the system.

2 **C**

- 3
- a No solutions. The regions are on opposite sides of the boundary line.
 - b Infinitely many solutions. The regions have the infinite set of points on the boundary line in common.

4

a $x < 1, y \leq 3$

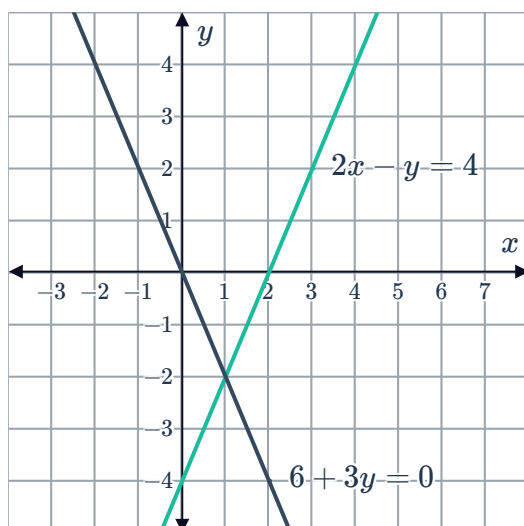
b $y \geq 2x - 3, y \leq 1$

c $x < 1, y < 1, y \geq -x - 5$

d $y + 3x < 0, y + 3x \leq 3$

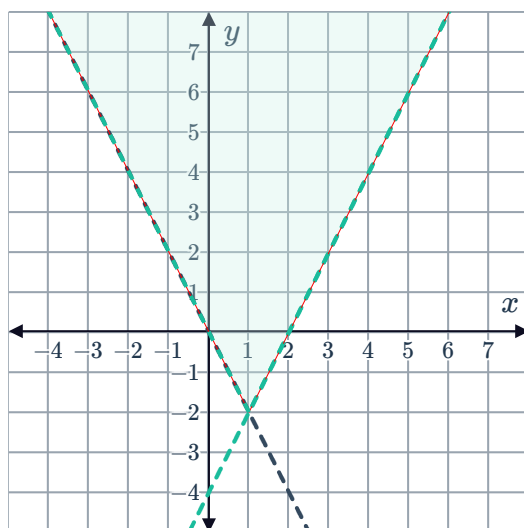
5

a

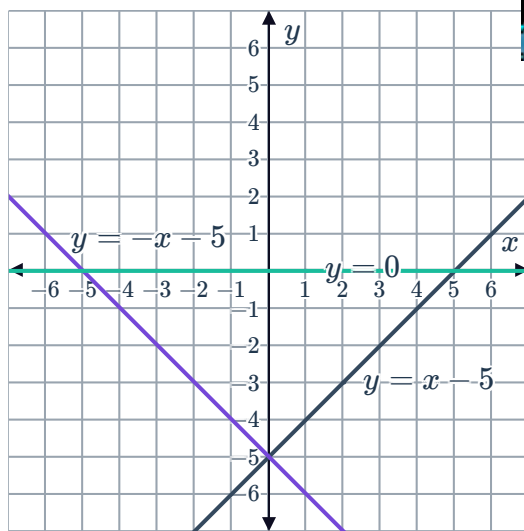


b Yes

c

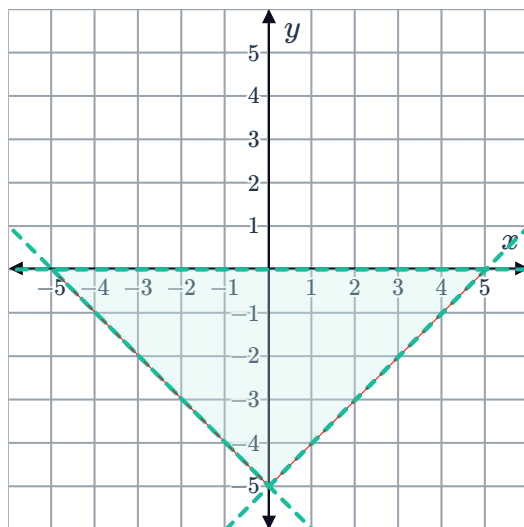


6 a



b Yes

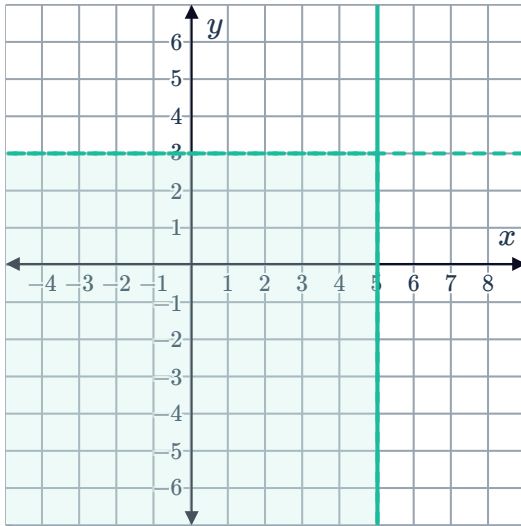
c



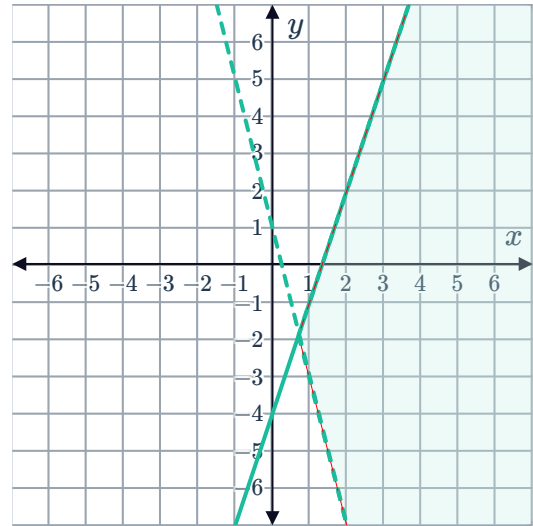
d 25 units^2

7

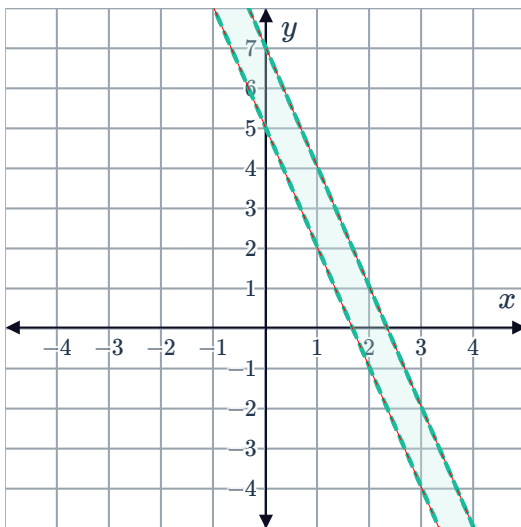
a



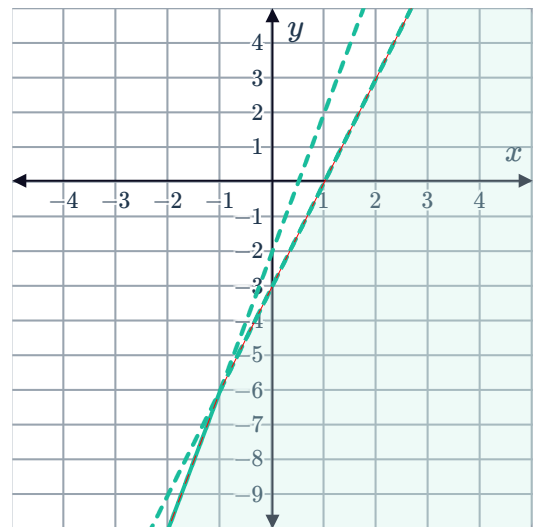
b



c

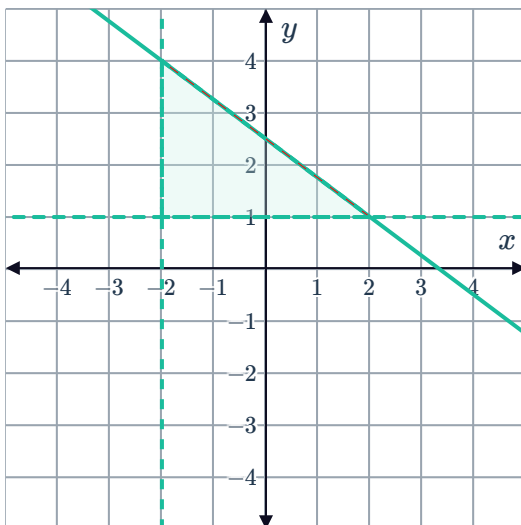


d



8

a



b 6 units²

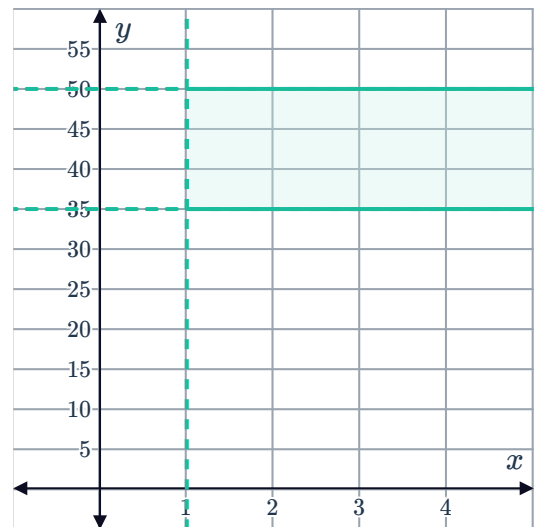


Applications

9

a $y \geq 35, y \leq 50$

b



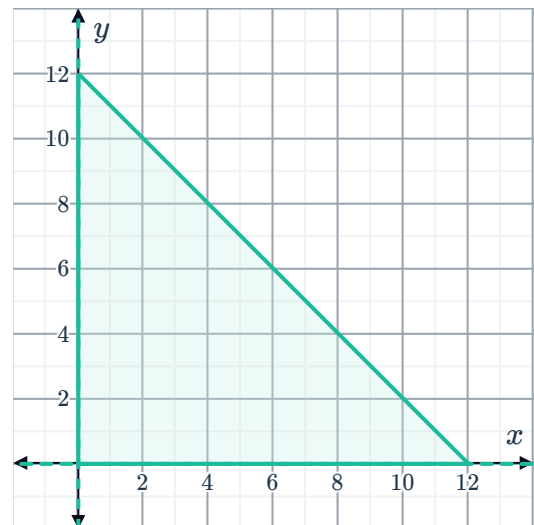
c Yes

d False

10

a $x \geq 0, 0 \leq y \leq 12 - x$

b



c
i Yes ii No iii No iv Yes

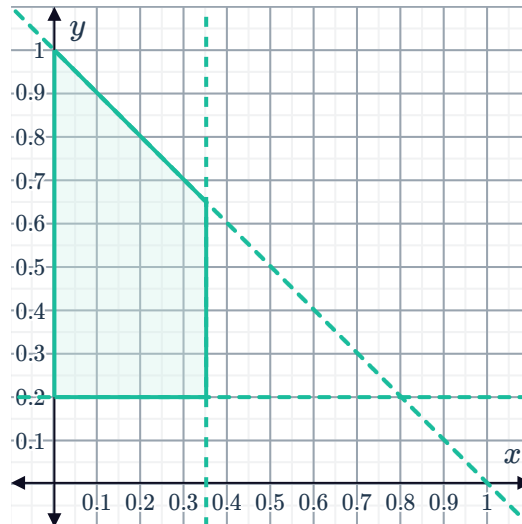
d $6 \text{ cm} \times 6 \text{ cm}$

11

a $0 \leq x \leq 0.35, y \geq 0.2, x + y \leq 1$



b



c

i Yes

ii No

iii Yes

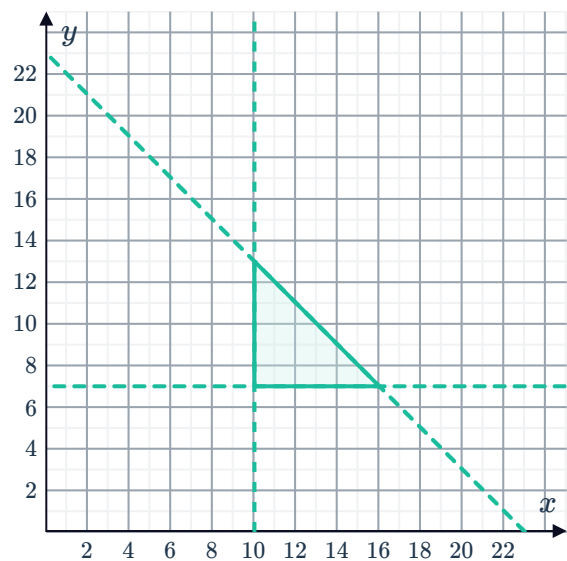
iv No

d Yes, because this would satisfy all of the inequalities and the point (0, 1) lies within the feasible region.

12

a $x \geq 10, y \geq 7, x + y \leq 23$

b



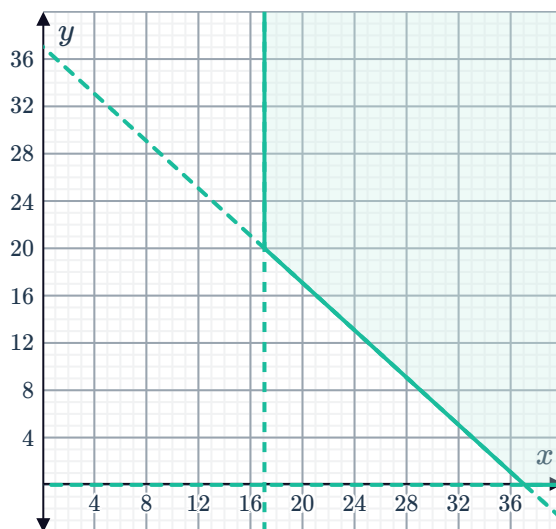
c 9

13

a $y \geq 0, x \geq 17, x + y \geq 37$



b



c

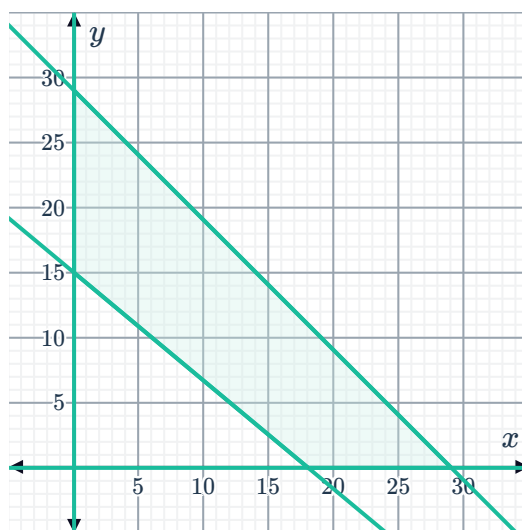
i No ii Yes iii No iv Yes

d 13

14

a $x \geq 0, y \geq 0, x + y \leq 29, 15x + 18y \geq 270$

b



c 10 hours

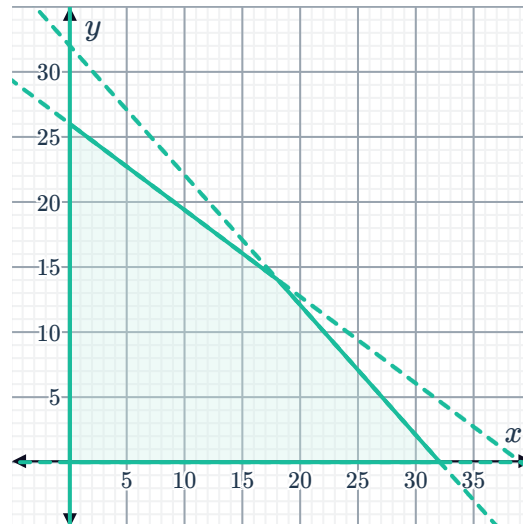
d Yes

15

a $x \geq 0, y \geq 0, y \leq 26 - \frac{2}{3}x$ and $y \leq 32 - x$



b



c A, B, C, D